
TP UBUNTU SOUS VIRTUAL BOX (Leo Fosseux)

Objectifs

Installer Ubuntu Desktop.

Découvrir et pratiquer les commandes Linux de base.

Manipuler des fichiers et répertoires simples.

Explorer les commandes réseau et gestion des processus

Première étape se rendre sur le site Ubuntu enfin lien qui à été communiquer par le professeur afin de télécharger l'iso :

<https://ubuntu.com/download/desktop>

Ubuntu 24.04.1 LTS



The latest LTS version of Ubuntu, for desktop PCs and laptops. LTS stands for long-term support — which means five years of free security and maintenance updates, extended up to 12 years with [Ubuntu Pro](#).

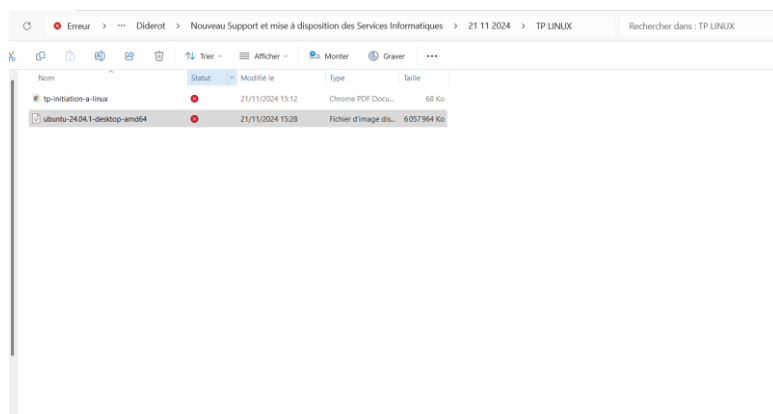
[Download 24.04.1 LTS](#) 5.8GB

For other versions of Ubuntu Desktop including torrents, the network installer, a list of local mirrors and past releases [check out our alternative downloads](#).

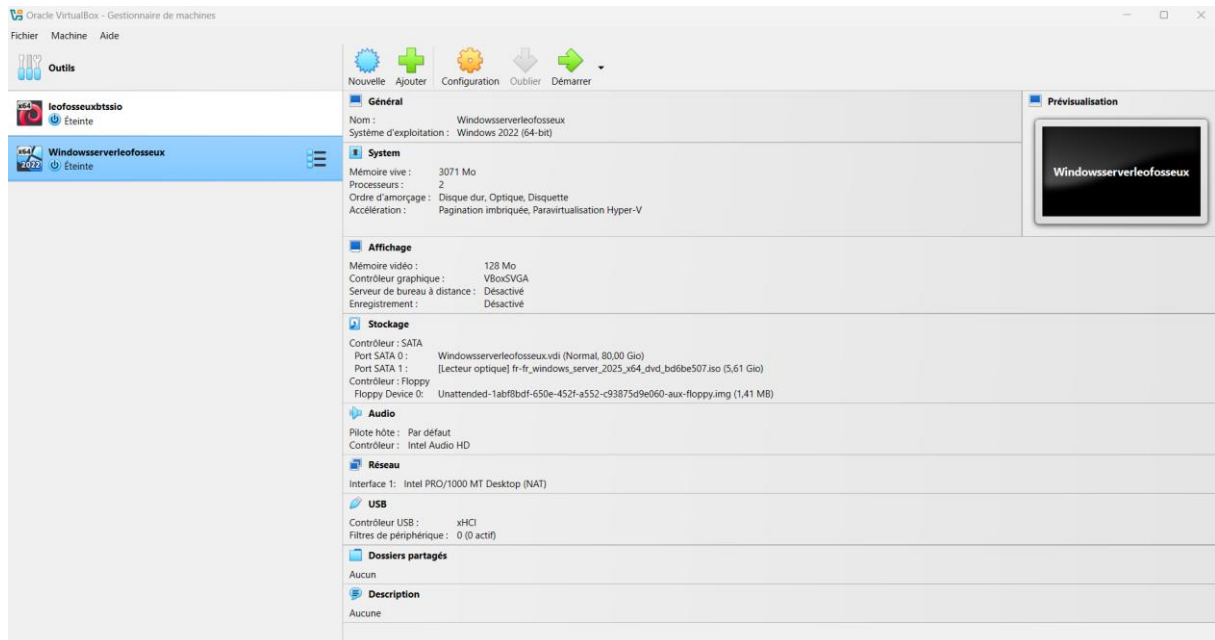
[What's new](#) [System requirements](#) [How to install](#)

- ✓ New Desktop installer with support for autoinstall
- ✓ New App Center and Firmware Updater applications
- ✓ GNOME 46 with support for quarter screen tiling
- ✓ Advanced Active Directory Group Policy Object support for Ubuntu Pro users

Deuxième étape : L'image ISO est enfin téléchargée et mis sur mon dossier TP LINUX :



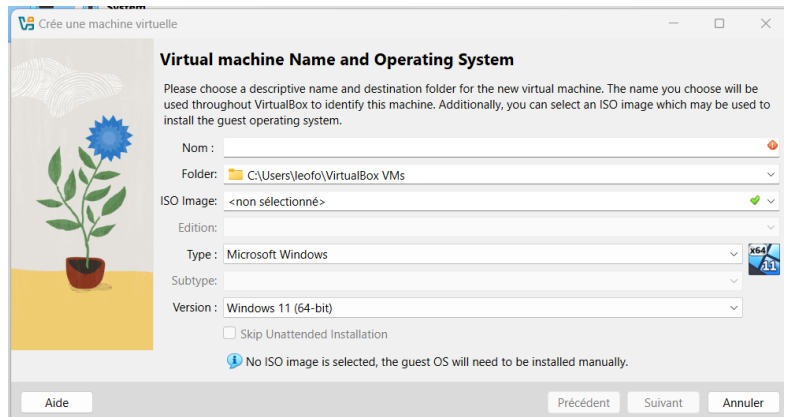
Troisième étape : Nous allons démarrer Virtual Box ! :



Quatrième étape : Maintenant, il faut cliquer sur nouvelle ! (La petite icone bleue)



Cinquième étape : Configuration de l'iso sous Virtual Box !



Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Nom :

Folder:

ISO Image:

Edition:

Type:

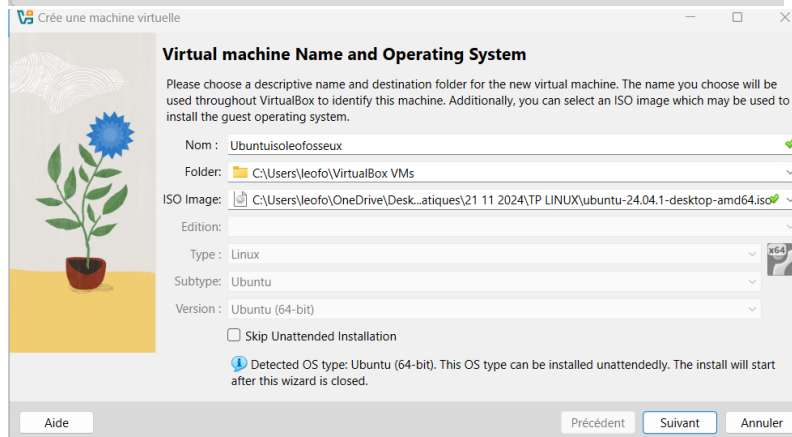
Subtype:

Version:

☐ Skip Unattended Installation

No ISO image is selected, the guest OS will need to be installed manually.

Aide Précédent Suivant Annuler



Virtual machine Name and Operating System

Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.

Nom:

Folder:

ISO Image:

Edition:

Type:

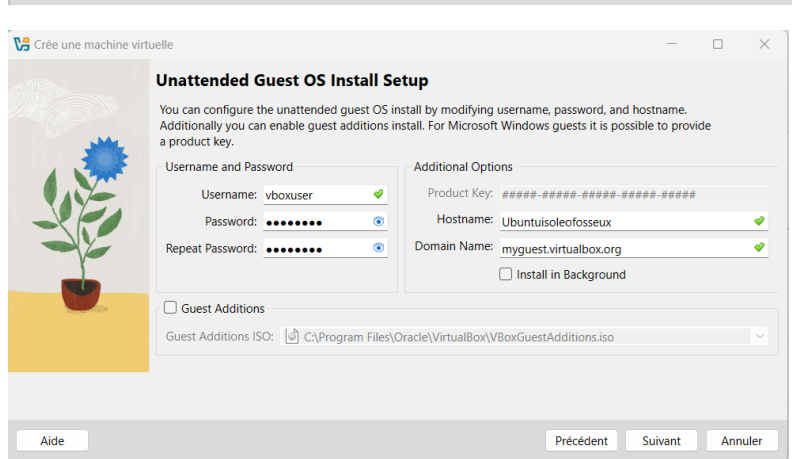
Subtype:

Version:

☐ Skip Unattended Installation

Detected OS type: Ubuntu (64-bit). This OS type can be installed unattendedly. The install will start after this wizard is closed.

Aide Précédent Suivant Annuler



Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username:

Password:

Repeat Password:

Additional Options

Product Key:

Hostname:

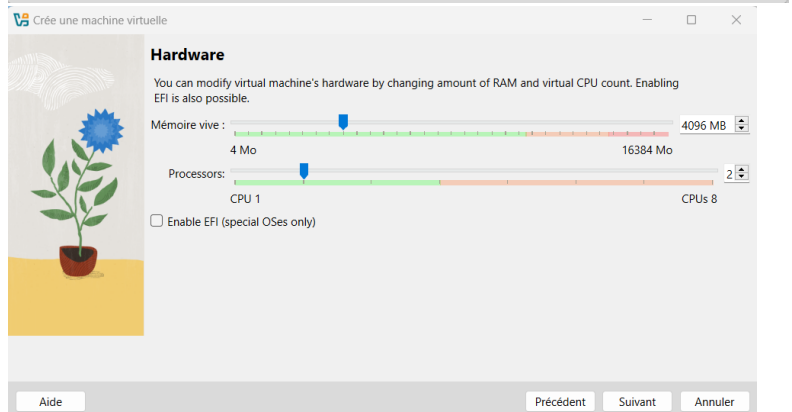
Domain Name:

☐ Install in Background

☐ Guest Additions

Guest Additions ISO:

Aide Précédent Suivant Annuler



Hardware

You can modify virtual machine's hardware by changing amount of RAM and virtual CPU count. Enabling EFI is also possible.

Mémoire vive :

Processors:

☐ Enable EFI (special OSes only)

Aide Précédent Suivant Annuler

Crée une machine virtuelle

Virtual Hard disk

If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.

☒ Create a Virtual Hard Disk Now

Disk Size: 20,00 Gio

☐ Pre-allocate Full Size

☐ Use an Existing Virtual Hard Disk File

Windowsserverleofosseux.vdi (Normal, 80,00 Gio)

☐ Do Not Add a Virtual Hard Disk

Aide Précédent Suivant Annuler

Crée une machine virtuelle

Récapitulatif

The following table summarizes the configuration you have chosen for the new virtual machine. When you are happy with the configuration press Finish to create the virtual machine. Alternatively you can go back and modify the configuration.

Machine Name and OS Type	
Machine Name	Ubuntuisoleofosseux
Machine Folder	C:/Users/leofo/VirtualBox VMs/Ubuntuisoleofosseux
ISO Image	C:/Users/leofo/OneDrive/Desktop/Leo/Diderot/Nouveau Support et mise à di...
Guest OS Type	Ubuntu (64-bit)
Skip Unattended Install	false

Unattended Install	
Username	vboxuser
Product Key	false
Hostname/Domain Name	Ubuntuisoleofosseux.myguest.virtualbox.org
Install in Background	false
Install Guest Additions	false

Hardware	
Mémoire vive	4096

Aide Précédent Finish Annuler

Général

Nom : Ubuntuisoleofosseux

Système d'exploitation : Ubuntu (64-bit)

System

Mémoire vive : 4096 Mo

Processeurs : 2

Ordre d'amorçage : Disque dur, Optique, Disquette

Accélération : Pagination imbriquée, Paravirtualisation KVM

Affichage

Mémoire vidéo : 16 Mo

Contrôleur graphique : VMSVGA

Serveur de bureau à distance : Désactivé

Enregistrement : Désactivé

Stockage

Contrôleur : IDE

Maître secondaire IDE : [Lecteur optique] Unattended-61a6b4b6-b108-4831-b678-8d4f910e7a64-aux-iso.viso (0 octets)

Contrôleur : SATA

Port SATA 0 : Ubuntuisoleofosseux.vdi (Normal, 20,00 Gio)

Audio

Pilote hôte : Par défaut

Contrôleur : ICH AC97

Réseau

Interface 1: Intel PRO/1000 MT Desktop (NAT)

USB

Contrôleur USB : OHCI, EHCI

Filtres de périphérique : 0 (0 actif)

Dossiers partagés

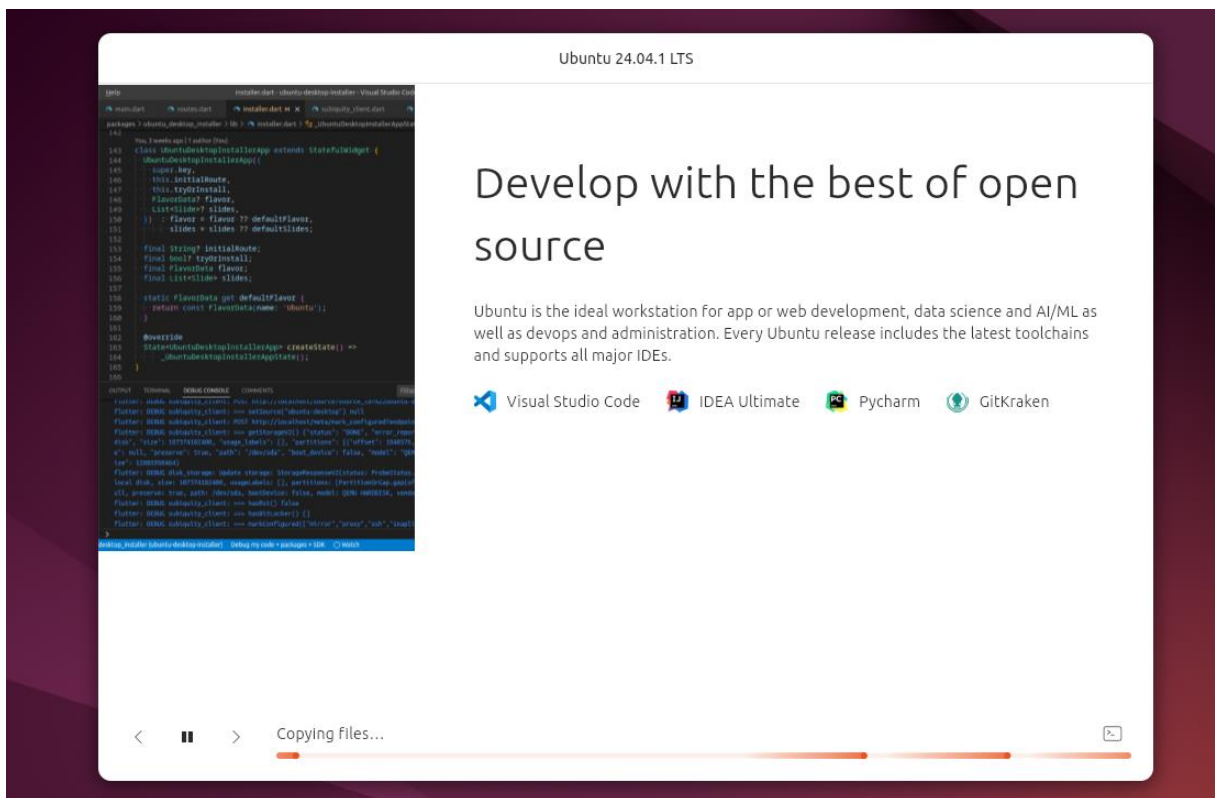
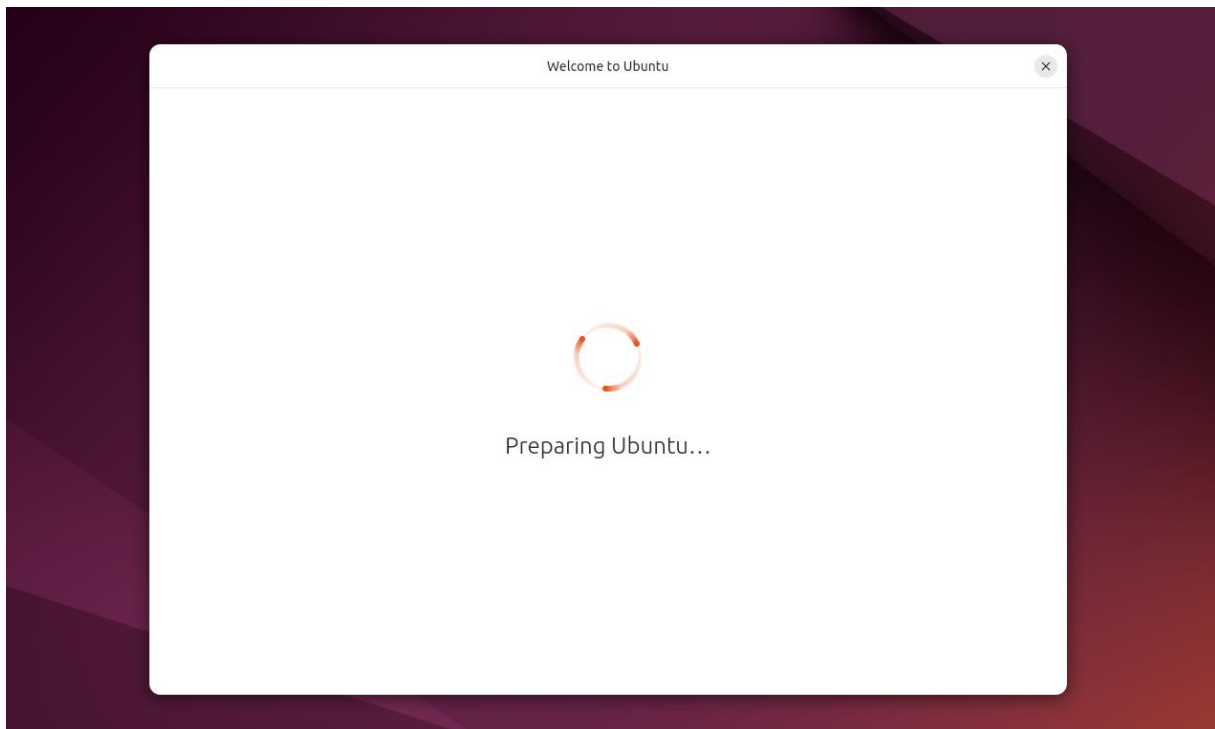
Aucun

Description

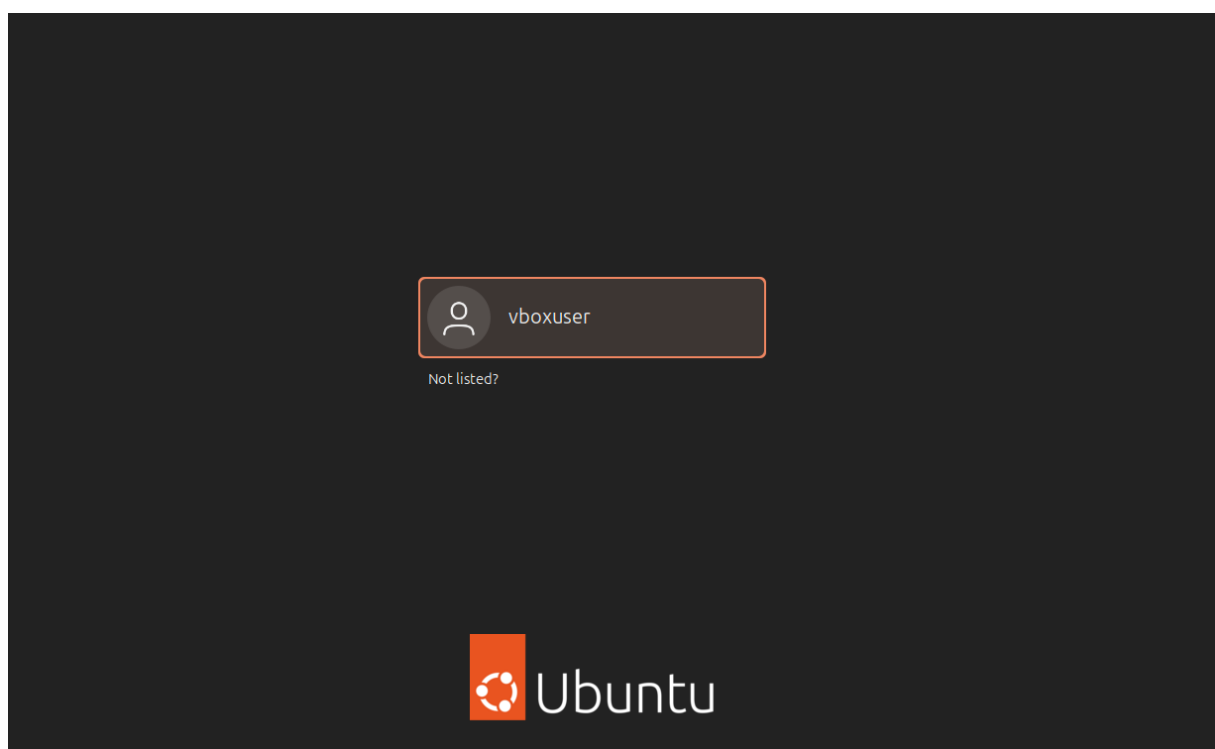
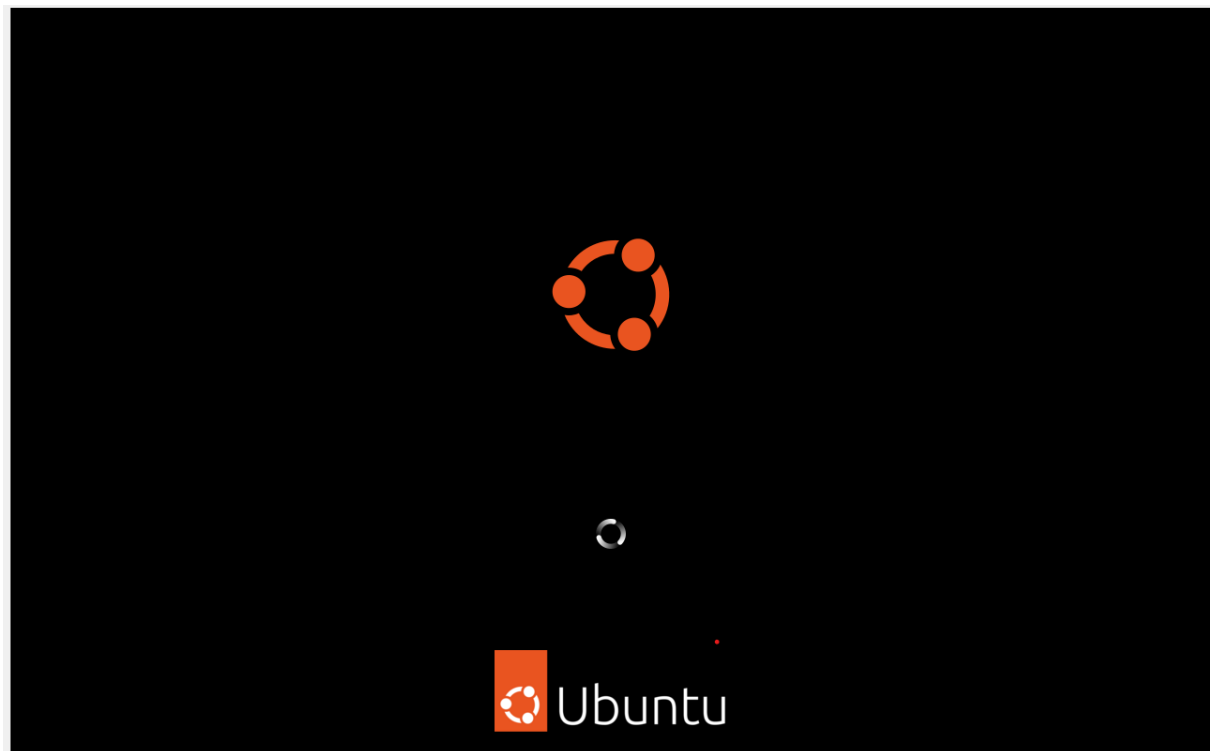
Aucune

La configuration de l'iso avec Virtual Box est enfin finis, Donc le fichier ISO est enfin lancer !

L'iso Ubuntu est entrain de se configurer



Redémarrage de l'iso après l'installation de Ubuntu :





L'installation de Ubuntu est enfin finalisée, donc Nous allons pouvoir enfin utiliser les commandes linux dans le terminal.

Dans le terminal, Les utilisateurs seront bloqués pour passer administrateurs afin d'exécuter les codes. J'ai donc trouvé la solution parce que sous Ubuntu, il n'y a pas de mot de passe « ROOT » !

```
vboxuser@Ubuntuisoleofosseux:~$ su root
Password:
su: Authentication failure
vboxuser@Ubuntuisoleofosseux:~$ sudo -i
[sudo] password for vboxuser:
root@Ubuntuisoleofosseux:~#
```

Nous allons commencer à exécuter les lignes de codes sous linux :

Créer un répertoire nommé "Diderot" : mkdir Diderot

```
root@Ubuntuisoleofosseux: ~  
See "man sudo_root" for details.  
  
vboxuser@Ubuntuisoleofosseux:~$ mkdir Diderot  
vboxuser@Ubuntuisoleofosseux:~$ su  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo  
usage: sudo -h | -K | -k | -V  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
           [-u user] [command [arg ...]]  
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]  
           [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
           [-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
           [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
           [-u user] file ...  
vboxuser@Ubuntuisoleofosseux:~$ su root  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo -i  
[sudo] password for vboxuser:  
root@Ubuntuisoleofosseux:~# mkdir Diderot  
root@Ubuntuisoleofosseux:~#
```


Créer un sous-répertoire "BTS_SIO" : mkdir Diderot/BTS_SIO

```
root@Ubuntuisoleofosseux: ~  
vboxuser@Ubuntuisoleofosseux:~$ mkdir Diderot  
vboxuser@Ubuntuisoleofosseux:~$ su  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo  
usage: sudo -h | -K | -k | -V  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
         [-u user] [command [arg ...]]  
usage: sudo [-ABbEHknPS] [-r role] [-t type] [-C num] [-D directory]  
         [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
         [-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
         [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
         [-u user] file ...  
vboxuser@Ubuntuisoleofosseux:~$ su root  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo -i  
[sudo] password for vboxuser:  
root@Ubuntuisoleofosseux:~# mkdir Diderot  
root@Ubuntuisoleofosseux:~# mkdir Diderot/BTS_SIO  
root@Ubuntuisoleofosseux:~#
```

Créer deux fichiers dans "BTS_SIO" : touch Diderot/BTS_SIO/Cours1.txt touch Diderot/BTS_SIO/TP1.txt

```
root@Ubuntuisoleofosseux: ~  
vboxuser@Ubuntuisoleofosseux:~$ su  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo  
usage: sudo -h | -K | -k | -V  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
         [-u user] [command [arg ...]]  
usage: sudo [-ABbEHknPS] [-r role] [-t type] [-C num] [-D directory]  
         [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
         [-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
         [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
         [-u user] file ...  
vboxuser@Ubuntuisoleofosseux:~$ su root  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo -i  
[sudo] password for vboxuser:  
root@Ubuntuisoleofosseux:~# mkdir Diderot  
root@Ubuntuisoleofosseux:~# mkdir Diderot/BTS_SIO  
root@Ubuntuisoleofosseux:~# touch Diderot/BTS_SIO/Cours1.txt  
touch Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~#
```

Lister le contenu du répertoire "BTS_SIO" : ls -l Diderot/BTS_SIO

```
root@Ubuntuisoleofosseux: ~  
usage: sudo -h | -K | -k | -V  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
          [-u user] [command [arg ...]]  
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]  
          [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
          [-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
          [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
          [-u user] file ...  
vboxuser@Ubuntuisoleofosseux:~$ su root  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo -i  
[sudo] password for vboxuser:  
root@Ubuntuisoleofosseux:~# mkdir Diderot  
root@Ubuntuisoleofosseux:~# mkdir Diderot/BTS_SIO  
root@Ubuntuisoleofosseux:~# touch Diderot/BTS_SIO/Cours1.txt  
touch Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~# ls -l Diderot/BTS_SIO  
total 0  
-rw-r--r-- 1 root root 0 Nov 21 15:31 Cours1.txt  
-rw-r--r-- 1 root root 0 Nov 21 15:31 TP1.txt  
root@Ubuntuisoleofosseux:~#
```

Supprimer le fichier "TP1.txt" : rm Diderot/BTS_SIO/TP1.txt

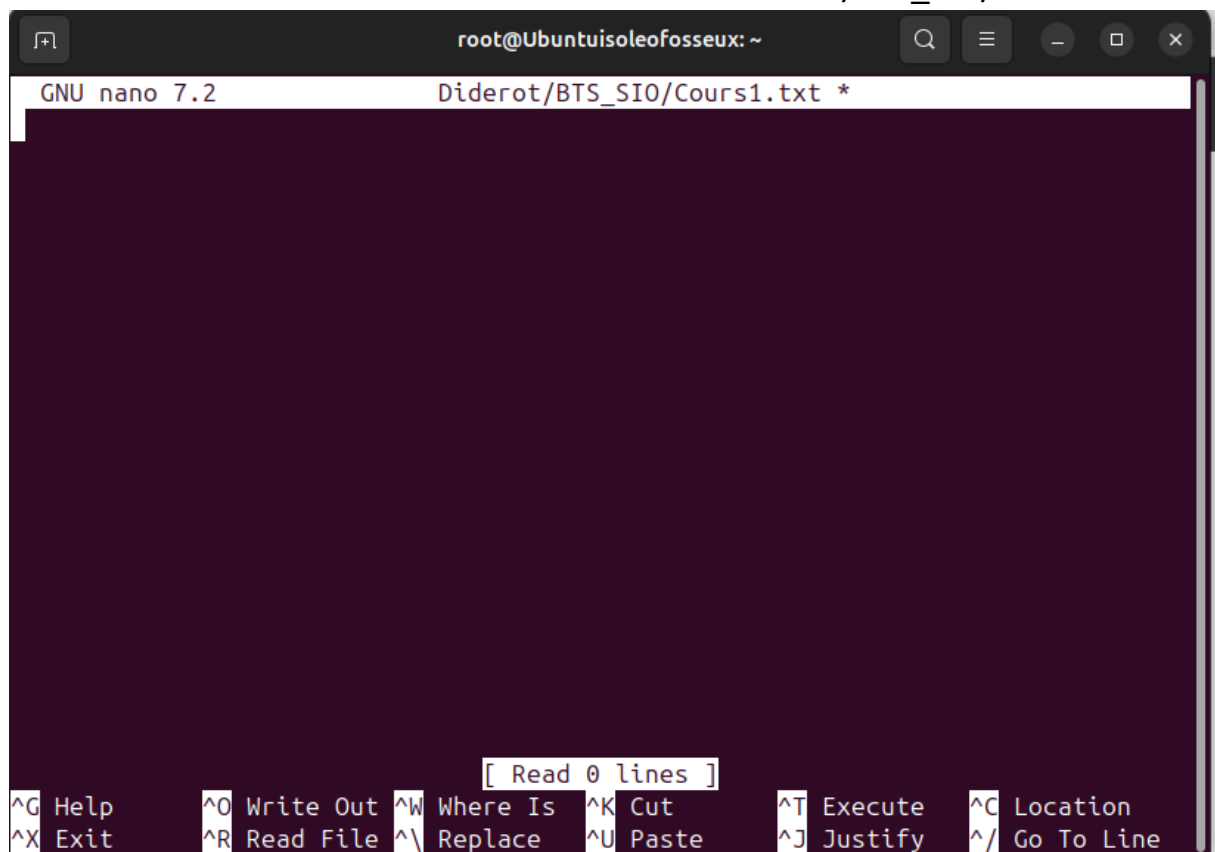
```
root@Ubuntuisoleofosseux: ~  
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]  
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]  
        [-u user] [command [arg ...]]  
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]  
        [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
        [-u user] [VAR=value] [-i | -s] [command [arg ...]]  
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]  
        [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]  
        [-u user] file ...  
vboxuser@Ubuntuisoleofosseux:~$ su root  
Password:  
su: Authentication failure  
vboxuser@Ubuntuisoleofosseux:~$ sudo -i  
[sudo] password for vboxuser:  
root@Ubuntuisoleofosseux:~# mkdir Diderot  
root@Ubuntuisoleofosseux:~# mkdir Diderot/BTS_SIO  
root@Ubuntuisoleofosseux:~# touch Diderot/BTS_SIO/Cours1.txt  
touch Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~# ls -l Diderot/BTS_SIO  
total 0  
-rw-r--r-- 1 root root 0 Nov 21 15:31 Cours1.txt  
-rw-r--r-- 1 root root 0 Nov 21 15:31 TP1.txt  
root@Ubuntuisoleofosseux:~# rm Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~#
```

Édition de fichiers :

Installer l'éditeur "nano" : sudo apt update && sudo apt install nano :

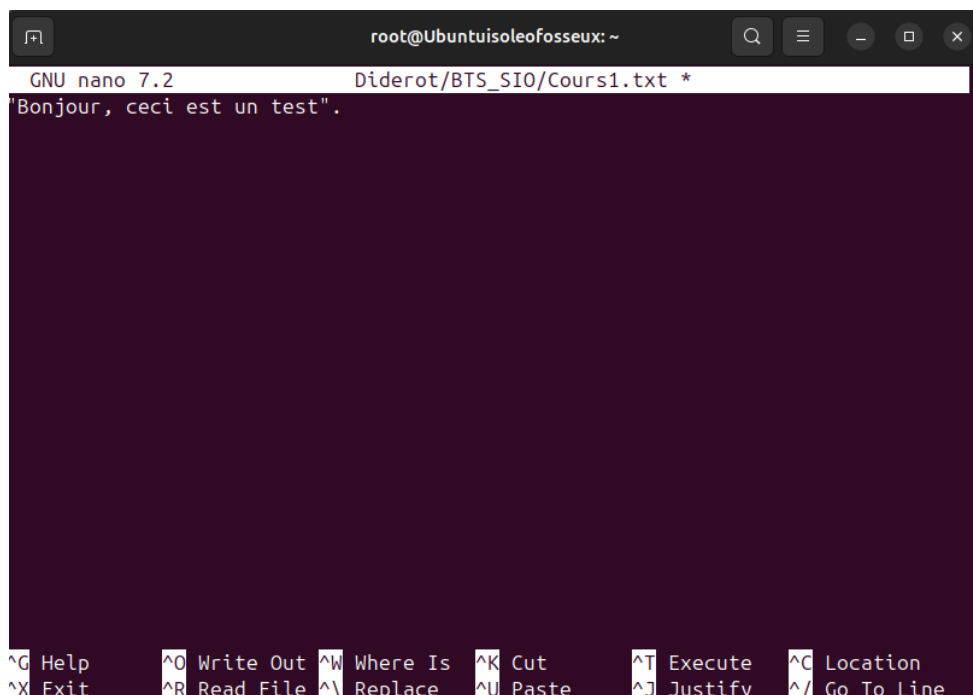
```
root@Ubuntuisoleofosseux:~# touch Diderot/BTS_SIO/Cours1.txt  
touch Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~# ls -l Diderot/BTS_SIO  
total 0  
-rw-r--r-- 1 root root 0 Nov 21 15:31 Cours1.txt  
-rw-r--r-- 1 root root 0 Nov 21 15:31 TP1.txt  
root@Ubuntuisoleofosseux:~# rm Diderot/BTS_SIO/TP1.txt  
root@Ubuntuisoleofosseux:~# sudo apt update && sudo apt install nano  
Hit:1 http://fr.archive.ubuntu.com/ubuntu noble InRelease  
Get:2 http://fr.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:4 http://fr.archive.ubuntu.com/ubuntu noble-backports InRelease  
Get:5 http://fr.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [670  
kB]  
Get:6 http://fr.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [157  
kB]  
Get:7 http://fr.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Package  
s [480 kB]  
Get:8 http://fr.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-e  
n [92.5 kB]  
Get:9 http://fr.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages  
[718 kB]  
Fetched 2,245 kB in 2s (905 kB/s)  
Reading package lists... 98%
```

Ouvrir et modifier le fichier "Cours1.txt" : nano Diderot/BTS_SIO/Cours1.txt



The screenshot shows a terminal window with the nano text editor open. The title bar indicates the user is root@Ubuntuisoleofosseux: ~. The nano header shows 'GNU nano 7.2' and the file path 'Diderot/BTS_SIO/Cours1.txt *'. The editor area is empty. The bottom status bar displays various keyboard shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, ^X Exit, ^R Read File, ^\ Replace, ^U Paste, ^J Justify, and ^_ Go To Line. A small status bar above the shortcuts shows '[Read 0 lines]'.

Dans l'éditeur, ajoutez une ligne de texte comme "Bonjour, ceci est un test".
Sauvegardez avec Ctrl+O et quittez avec Ctrl+X.



The screenshot shows the same terminal window with the nano text editor. The file 'Diderot/BTS_SIO/Cours1.txt' now contains the text 'Bonjour, ceci est un test.'. The status bar and keyboard shortcuts remain the same as in the previous screenshot.

Le fichier est maintenant enregistré et nous avons pu quitter avec la touche control + X !

Afficher le contenu du fichier "Cours1.txt" : cat Diderot/BTS_SIO/Cours1.txt

```
root@Ubuntuisoleofosseux: ~  
kB]  
Get:6 http://fr.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [157  
kB]  
Get:7 http://fr.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Package  
s [480 kB]  
Get:8 http://fr.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-e  
n [92.5 kB]  
Get:9 http://fr.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages  
[718 kB]  
Fetched 2,245 kB in 2s (905 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
90 packages can be upgraded. Run 'apt list --upgradable' to see them.  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
nano is already the newest version (7.2-2ubuntu0.1).  
nano set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 90 not upgraded.  
root@Ubuntuisoleofosseux:~# nano Diderot/BTS_SIO/Cours1.txt  
root@Ubuntuisoleofosseux:~# cat Diderot/BTS_SIO/Cours1.txt  
"Bonjour, ceci est un test".  
root@Ubuntuisoleofosseux:~#
```

3. Commandes Réseau ! :

1. Tester la connectivité réseau avec un ping vers "google.com" : ping -c 4

```
root@Ubuntuisoleofosseux: ~  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
nano is already the newest version (7.2-2ubuntu0.1).  
nano set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 90 not upgraded.  
root@Ubuntuisoleofosseux:~# nano Diderot/BTS_SIO/Cours1.txt  
root@Ubuntuisoleofosseux:~# cat Diderot/BTS_SIO/Cours1.txt  
"Bonjour, ceci est un test".  
root@Ubuntuisoleofosseux:~# ping -c 4 google.com  
PING google.com (172.217.18.206) 56(84) bytes of data:  
64 bytes from ham02s14-in-f206.1e100.net (172.217.18.206): icmp_seq=1 ttl=255 ti  
me=27.9 ms  
64 bytes from ham02s14-in-f206.1e100.net (172.217.18.206): icmp_seq=2 ttl=255 ti  
me=27.6 ms  
64 bytes from ham02s14-in-f206.1e100.net (172.217.18.206): icmp_seq=3 ttl=255 ti  
me=26.5 ms  
64 bytes from ham02s14-in-f206.1e100.net (172.217.18.206): icmp_seq=4 ttl=255 ti  
me=27.7 ms  
  
--- google.com ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3004ms  
rtt min/avg/max/mdev = 26.500/27.428/27.901/0.548 ms  
root@Ubuntuisoleofosseux:~#
```

4. Gestion des Processus ! :

Afficher la liste des processus en cours : ps aux

```
root@Ubuntuisoleofosseux: ~  
vboxuser 4224 0.0 1.2 208020 48896 ? Sl 15:14 0:00 /snap/firefox  
vboxuser 4275 0.0 2.3 2423932 94536 ? Sl 15:14 0:01 /snap/firefox  
vboxuser 4304 0.1 3.3 2451612 132984 ? Sl 15:14 0:03 /snap/firefox  
vboxuser 4346 0.0 0.4 1760984 19840 ? Sl 15:14 0:00 /usr/bin/snap  
vboxuser 4678 4.4 13.4 2994072 540080 ? Sl 15:14 1:21 /snap/firefox  
vboxuser 4849 0.0 1.4 341836 57600 ? Sl 15:14 0:00 /snap/firefox  
vboxuser 4860 0.2 2.4 2414248 96260 ? Sl 15:14 0:04 /snap/firefox  
root 4933 0.0 0.0 0 0 ? I< 15:15 0:00 [kworker/1:0H  
vboxuser 4962 0.0 1.8 2388928 73344 ? Sl 15:15 0:00 /snap/firefox  
vboxuser 5010 0.0 1.8 2388928 73088 ? Sl 15:15 0:00 /snap/firefox  
vboxuser 5043 0.0 1.3 340552 55680 ? Sl 15:16 0:00 /snap/firefox  
vboxuser 5091 0.0 1.8 2388908 73336 ? Sl 15:16 0:00 /snap/firefox  
vboxuser 5227 0.2 1.3 706440 56008 ? Ssl 15:17 0:03 /usr/libexec/  
vboxuser 5243 0.0 0.1 19692 4992 pts/0 Ss 15:17 0:00 bash  
root 5433 0.0 0.1 28480 7936 pts/0 S+ 15:22 0:00 sudo -i  
root 5440 0.0 0.0 28480 2624 pts/1 Ss 15:22 0:00 sudo -i  
root 5441 0.0 0.1 19680 5120 pts/1 S 15:22 0:00 -bash  
root 5536 0.1 0.0 0 0 ? I 15:28 0:01 [kworker/u4:1  
root 5554 0.0 0.0 0 0 ? I 15:30 0:00 [kworker/0:0]  
root 5631 0.0 0.0 0 0 ? I 15:36 0:00 [kworker/u4:2  
root 5638 0.1 0.0 0 0 ? I 15:36 0:00 [kworker/u4:3  
root 5664 0.0 0.0 0 0 ? I 15:37 0:00 [kworker/1:2-  
root 6186 200 0.1 22416 4736 pts/1 R+ 15:45 0:00 ps aux  
root@Ubuntuisoleofosseux:~#
```

Surveiller les processus en temps réel : top

```
root@Ubuntuisoleofosseux: ~  
top - 15:46:19 up 41 min, 1 user, load average: 0.37, 0.29, 0.37  
Tasks: 197 total, 2 running, 195 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 5.0 us, 7.9 sy, 0.0 ni, 86.0 id, 0.0 wa, 0.0 hi, 1.2 si, 0.0 st  
MiB Mem : 3916.1 total, 321.6 free, 2007.3 used, 1883.9 buff/cache  
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 1908.8 avail Mem  


| PID  | USER     | PR | NI  | VIRT    | RES    | SHR    | S | %CPU | %MEM | TIME+   | COMMAND  |
|------|----------|----|-----|---------|--------|--------|---|------|------|---------|----------|
| 2462 | vboxuser | 20 | 0   | 4036836 | 415852 | 154172 | S | 21.9 | 10.4 | 6:02.50 | gnome-s+ |
| 4020 | vboxuser | 20 | 0   | 11.3g   | 547308 | 223936 | R | 10.3 | 13.6 | 2:42.70 | firefox  |
| 4678 | vboxuser | 20 | 0   | 2994056 | 542188 | 119080 | S | 7.3  | 13.5 | 1:24.29 | Isolate+ |
| 5536 | root     | 20 | 0   | 0       | 0      | 0      | I | 1.0  | 0.0  | 0:01.53 | kworker+ |
| 4860 | vboxuser | 20 | 0   | 2414264 | 96328  | 72180  | S | 0.7  | 2.4  | 0:05.12 | Isolate+ |
| 5227 | vboxuser | 20 | 0   | 706440  | 55852  | 44296  | S | 0.7  | 1.4  | 0:03.71 | gnome-t+ |
| 4962 | vboxuser | 20 | 0   | 2388928 | 73344  | 60928  | S | 0.3  | 1.8  | 0:00.74 | Web Con+ |
| 1    | root     | 20 | 0   | 22960   | 14040  | 9432   | S | 0.0  | 0.4  | 0:03.91 | systemd  |
| 2    | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.01 | kthreadd |
| 3    | root     | 20 | 0   | 0       | 0      | 0      | S | 0.0  | 0.0  | 0:00.00 | pool_wo  |
| 4    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |
| 5    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |
| 6    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |
| 7    | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |
| 9    | root     | 20 | 0   | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:01.60 | kworker+ |
| 10   | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |
| 12   | root     | 0  | -20 | 0       | 0      | 0      | I | 0.0  | 0.0  | 0:00.00 | kworker+ |


```

Conclusion de ce TP :

Je n'ai ressenti aucune complication de mon côté pour réaliser ce TP sous linux.

Les étapes données par M. BENCHEIKH ont été très clairs pour moi.

J'ai ressenti du plaisir de manipuler et mettre en avant mes compétences de savoir-faire en informatique.